Socially Optimal Attention

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- *Attentional externalities* $\Rightarrow$ socially optimal $\neq$ privately optimal

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- Emphasis here: examples, intuition

Set up

Standard assumptions:

- Subjective perception of variable x is x^s, where attention $m \in [0,1]$ weights true value x vs. prior x^d (note Bayesian connection):

$$x^s = mx + (1-m)x^d$$

- At $m = 1$, fully rational; at $m = 0$ no reaction and belief = prior
- Utility $u(a(x^s), x)$, $u_a > 0$, $u_{aa} < 0$. Choose action $a(x^s)$
- x impacts action and level of u (e.g. x = price of bananas, $a(x^s)$ = demand)
- Attention is costly: $\mathcal{C}(m) = \kappa m$

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New assumptions:

- Unit continuum of identical individuals $\Rightarrow$ single individual's attention cannot impact x
- Social planner is choosing attention for whole population, so $x(m)$ — SP takes into account externality

Privately optimal attention: Gabaix (2014), part 1/2

1. **Problem**: individual wants to minimize losses from inattention and costs of attention :

$$\max_{m \in [0,1]} u(m) - u(1) - \kappa m$$

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$$u(m) - u(1) \approx u_m(a(x^s(m)), x)(m-1) + \frac{1}{2} u_{mm}(a(x^s(m)), x) (m-1)^2$$

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- Goal: maximize approx. ℓosses from inattention $-$ costs of attention

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- More attention when more sensitive to price ($|D'(p)|$ high) and when prices change often (σ_p^2 high)

This paper: socially optimal attention, part 1/2

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Where $\zeta \equiv u_{xx}(x_m)^2 + u_x x_{mm}$ captures second-order impacts of externality

Socially optimal attention, part 2/2

3. **Soluton**:

$$m^*_{\text{social, exact}} = 1 + \frac{\kappa - u_x x_m}{\ell - \ell \cdot \left(\frac{x_m}{x - x^d}\right)^2 + \zeta}$$

- Hard to interpret, so linear approx. in limit of small externality (i.e. in x_m around $x_m = 0$), yields

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- Relative size of m^*_{social} hinges entirely on sign of dp^*/dm — will more attention to price increase or decrease the monopolist's price?

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- Optimal price follows:

Behavioral inverse elasticity rule

$$p^* = c - \frac{D(p^s)}{D'(p^s)} \cdot \frac{1}{m}$$

Some behavioral IO: how price responds to attention

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- **Final effect**: price can *increase or decrease* with attention ▸ General expression
  - Depends on prior p^d and log-curvature of demand ▸ CES, logit, exp
  - Suff. stats: p^d and cost pass-through dp^*/dc or prior (p^d) pass-through dp^*/dp^d ▸ More

Monopoly price and attention: linear demand

- For intuition, linear demand: $D(p^s) = A - \alpha p^s$
- Given $\pi = (p - c) \cdot \left(A - \alpha \left(mp + (1 - m)p^d\right)\right)$, the optimal price is

$$p^* = \frac{A - \alpha p^d}{2\alpha m} + \frac{p^d + c}{2}$$

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- Increasing attention $\Rightarrow$ *less influence* of p^d
- If high p^d pushes price down, then high attention m decreases role of p^d, and thus of downward pressure

Socially optimal attention: linear demand

- Keeping linear demand corresponds to quadratic utility:

$$u(Q, p^s) = \frac{AQ}{\alpha} - \frac{Q^2}{2\alpha} - pQ$$

Then $Q^* = D(p^s) = A - \alpha p^s$, and we have

- **Privately optimal attention**:

$$m^*_{\text{private}} = 1 - \frac{\kappa}{\alpha \cdot \sigma_p^2}$$

- **Socially optimal attention**:

$$\begin{aligned} m^*_{\text{social}} &= m^*_{\text{private}} - \frac{dp^*}{dm} \cdot \frac{D(p^*)}{\alpha\sigma_p^2} \\ &= m^*_{\text{private}} + \left(A - \alpha p^d\right) \cdot \frac{D(p^*)}{2\alpha^2\sigma_p^2} \end{aligned}$$

Conclusion

- New definition of optimal attention: SP
- Size/sign of externality can be measured using suff. stats
- Notion of m^*_{social} is more general, e.g. attention of firms to each other's prices
- Shows that attention isn't always good!
- Limitations: no heterogeneity, no dynamics, local analysis

Thought experiment

In which state of the world is utility higher?

1. You're inattentive to price, so you buy a suboptimal number of bananas, but they are cheap
2. You're fully attentive to price, so you buy exactly your optimal number of bananas, but they are expensive

In case (1), you face utility losses from consuming wrong amount, but you paid a low price (which is a 'good' state of the world). In case (2), you bought exactly the right number of bananas, but they were expensive.

Thought experiment, cont.

	Inattentive	**Attentive**
High price	$p^s = 1,\ p = 3$	$p^s = p = 3$
Low price	$p^s = 1,\ p = 2$	$p^s = p = 2$

Table: Beliefs and states of the world

- Suppose utility is $u(c; p) = \ln(c) - c \cdot p + y$, for $y = 5$ and we have that wrong beliefs $\implies$ wrong actions $\implies$ misoptimization.
- So, $c^* = 1/p^s$, and so experienced utility is $u(c^*(p^s); p)$, i.e. consumption based on perceived price but still paying true price in the end.

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	Inattentive	**Attentive**
High price	$u_{p^s=1,\ p=3} = 2$	$u_{p^s=p=3} = 2.9$
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Low price	$u_{p^s=1,\ p=2} = 3$	$u_{p^s=p=2} = 3.31$

Table: Payoffs

Behavioral inverse elasticity rule: alternative expression

This may look straightforward w.r.t. m, but it's not

$$p^* = c - \frac{D(p^s)}{D'(p^s)} \cdot \frac{1}{m}$$

Here is an equivalent expression, written in terms of elasticities:

$$p^* = c \cdot \frac{\epsilon^s}{\epsilon^s + 1} - \left(\frac{1-m}{m}\right) \cdot \frac{p^d}{\epsilon^s + 1}$$

And compare to traditional expression (i.e. $m = 1$):

$$p^* = c \cdot \frac{\epsilon}{\epsilon + 1}$$

How price changes with attention

Attentional pass-through

$$\begin{aligned}\frac{dp^*}{dm} &= \frac{1}{m\left(1-\mu'\right)}\left[\mu'\left(p^* - p^d\right) - \frac{\mu(p^s)}{m}\right] \\ &= \frac{\mu'}{1-\mu'}\cdot\left(\frac{c-p^d}{m}\right) - \frac{\mu(p^s)}{m^2}\end{aligned}$$

More examples of dp^*/dm

1. CES demand: $D = (p^s)^{-\psi}$, then

$$p^* = \underbrace{\frac{\psi c}{\psi - 1}}_{=p^r} + \frac{(1-m)\,p^d}{m\,(\psi-1)} \implies$$

$$\frac{dp^*}{dm} = -\frac{p^d}{m(\psi-1)} - \frac{(1-m)p^d}{m^2(\psi-1)} < 0$$

2. Logit demand: $D_1(p^s) = \frac{\exp(q-p^s)}{\exp(q-p^s)+1}$, then

$$p^* = c + \frac{1}{m\,(1-D(p^s))} \implies$$

$$\frac{dp^*}{dm} = \frac{mD(p^s)\cdot(1-(D(p^s)))\cdot\left(p^d-c\right)-1}{m^2\cdot(1-D(p^s))} \lessgtr 0$$

3. Exponential demand: $D = \exp(-p^s)$, then

Sufficient statistics for dp^*/dm

1. **Cost pass-through**: Attentional pass-through can be expressed using $\rho \equiv dp^*/dc$:

$$\frac{dp^*}{dm} = \frac{\mu}{m^2}\left[(1-\rho)\cdot\left(\frac{p^d - c}{p^* - c}\right) - 1\right]$$

2. **Prior pass-through**: Attentional pass-through can be expressed using dp^*/dp^d:

$$\frac{dp^*}{dm} = \frac{dp^*}{dp^d}\cdot\frac{c - p^d}{1-m} - \frac{\mu}{m^2}$$

3. Note that the above two are related as

$$\frac{dp^*}{dp^d} = \rho\mu'\cdot\frac{1-m}{m}$$

Gabaix, Xavier. 2014. "A sparsity-based model of bounded rationality." *Quarterly Journal of Economics*, 129: 1661–1710.